

15.8 Solution: For multi-particle collision, we can generalize Eq. (15.1) for two identical particles with charge q as

$$\frac{d^2 N}{d(\hbar\omega)d\Omega} = \frac{q^2}{\hbar c} \frac{1}{4\pi^2} \frac{1}{\hbar\omega} \left| \sum_{i=1}^2 \int \frac{d}{dt} \left[\frac{\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}_i)}{1 - \mathbf{n} \cdot \boldsymbol{\beta}_i} \right] \exp \left(i\omega \left(t - \frac{\mathbf{n} \cdot \mathbf{r}_i(t)}{c} \right) \right) dt \right|^2.$$

In the c.m.s. system, we have

$$\boldsymbol{\beta}_1 = -\boldsymbol{\beta}_2 = \boldsymbol{\beta}' = \frac{\boldsymbol{\beta}}{2}, \quad \mathbf{r}_1(t) = -\mathbf{r}_2(t) = \frac{\mathbf{r}(t)}{2},$$

and

$$\begin{aligned} \frac{d^2 N}{d(\hbar\omega)d\Omega} &= \frac{q^2}{\hbar c} \frac{1}{4\pi^2} \frac{1}{\hbar\omega} \left| \int \frac{d}{dt} \left[\frac{\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}')}{1 - \mathbf{n} \cdot \boldsymbol{\beta}'} \right] \exp \left(i\omega \left(t - \frac{\mathbf{n} \cdot \mathbf{r}(t)}{2c} \right) \right) dt \right. \\ &\quad \left. - \int \frac{d}{dt} \left[\frac{\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}')}{1 + \mathbf{n} \cdot \boldsymbol{\beta}'} \right] \exp \left(i\omega \left(t + \frac{\mathbf{n} \cdot \mathbf{r}(t)}{2c} \right) \right) dt \right|^2. \end{aligned}$$

From here, we will stick with $\boldsymbol{\beta}'$ and will replace it with the appropriate value in the last step.

To obtain the lowest nonvanishing approximation to the radiation cross section, assume $\beta \ll 1$ and $\omega R/c \ll 1$ and expand, the radiation cross section becomes

$$\begin{aligned} \frac{d^2 N}{d(\hbar\omega)d\Omega} &= \frac{q^2}{\hbar c} \frac{1}{4\pi^2} \frac{1}{\hbar\omega} \left| \int \frac{d}{dt} \left[(\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}')) (1 + \mathbf{n} \cdot \boldsymbol{\beta}') \right] \left(1 - \frac{i\omega}{2c} \mathbf{n} \cdot \mathbf{r}(t) \right) e^{i\omega t} dt \right. \\ &\quad \left. - \int \frac{d}{dt} \left[(\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}')) (1 - \mathbf{n} \cdot \boldsymbol{\beta}') \right] \left(1 + \frac{i\omega}{2c} \mathbf{n} \cdot \mathbf{r}(t) \right) e^{i\omega t} dt \right|^2 \\ &= \frac{q^2}{\hbar c} \frac{1}{\pi^2} \frac{1}{\hbar\omega} \left| \int \frac{d}{dt} \left[(\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}')) (\mathbf{n} \cdot \boldsymbol{\beta}') \right] e^{i\omega t} dt \right. \\ &\quad \left. - \frac{i\omega}{2c} \int \frac{d}{dt} (\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}')) (\mathbf{n} \cdot \mathbf{r}(t)) e^{i\omega t} dt \right|^2. \end{aligned}$$

Ignoring electromagnetic interaction, the collision can be viewed as completed in a very short time. This allows us to drop the $e^{i\omega t}$ term and obtain

$$\begin{aligned} \frac{d^2 N}{d(\hbar\omega)d\Omega} &= \frac{q^2}{\hbar c} \frac{1}{\pi^2} \frac{1}{\hbar\omega} \left| (\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}'_f)) (\mathbf{n} \cdot \boldsymbol{\beta}'_f) - (\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}'_i)) (\mathbf{n} \cdot \boldsymbol{\beta}'_i) \right. \\ &\quad \left. - \frac{i\omega}{2c} (\mathbf{n} \times (\mathbf{n} \times (\boldsymbol{\beta}'_f - \boldsymbol{\beta}'_i))) (\mathbf{n} \cdot \mathbf{r}) \right|^2, \end{aligned}$$

where $\mathbf{r}$ is the point of impact on the hard sphere. We see that the first line and the second line have a phase difference of $\pi/2$ and therefore can be calculated separately.

(i) For the first part, we can perform the calculation in exactly the same way as in Problem 15.3. For the incoming part, $\boldsymbol{\beta}'_i = \boldsymbol{\beta}'(0, 0, 1)$. Also,

$$\mathbf{n} = (\sin \theta, 0, \cos \theta), \quad \boldsymbol{\epsilon}_1 = (\cos \theta, 0, -\sin \theta), \quad \boldsymbol{\epsilon}_2 = (0, 1, 0),$$

then

$$\epsilon_1 \cdot \left(\mathbf{n} \times (\mathbf{n} \times \beta'_i) \right) (\mathbf{n} \cdot \beta'_i) = -(\epsilon_1 \cdot \beta'_i) (\mathbf{n} \cdot \beta'_i) = \beta'^2 \sin \theta \cos \theta,$$

and

$$\epsilon_2 \cdot \left(\mathbf{n} \times (\mathbf{n} \times \beta'_i) \right) (\mathbf{n} \cdot \beta'_i) = -(\epsilon_2 \cdot \beta'_i) (\mathbf{n} \cdot \beta'_i) = 0.$$

For the outgoing part, assume that $\mathbf{n}$ is in the z -direction and β'_f makes an angle of θ_p with it, as in Problem 15.3. Then,

$$\beta'_f = \beta'(\sin \theta_p \cos \phi_p, \sin \theta_p \sin \phi_p, \cos \theta_p), \quad \mathbf{n} = (0, 0, 1), \quad \epsilon_1 = (1, 0, 0), \quad \epsilon_2 = (0, 1, 0),$$

and

$$\epsilon_1 \cdot \left(\mathbf{n} \times (\mathbf{n} \times \beta'_f) \right) (\mathbf{n} \cdot \beta'_f) = -(\epsilon_1 \cdot \beta'_f) (\mathbf{n} \cdot \beta'_f) = -\beta'^2 \sin \theta_p \cos \theta_p \cos \phi_p,$$

$$\epsilon_2 \cdot \left(\mathbf{n} \times (\mathbf{n} \times \beta'_f) \right) (\mathbf{n} \cdot \beta'_f) = -(\epsilon_2 \cdot \beta'_f) (\mathbf{n} \cdot \beta'_f) = -\beta'^2 \sin \theta_p \cos \theta_p \sin \phi_p.$$

Therefore, the contribution from the first part is

$$\begin{aligned} & \sum_{i=1}^2 \left| \epsilon_i \cdot \left[\left(\mathbf{n} \times (\mathbf{n} \times \beta'_f) \right) (\mathbf{n} \cdot \beta'_f) - \left(\mathbf{n} \times (\mathbf{n} \times \beta'_i) \right) (\mathbf{n} \cdot \beta'_i) \right] \right|^2 \\ &= \beta'^4 \left((\sin \theta_p \cos \theta_p \cos \phi_p + \sin \theta \cos \theta)^2 + \sin^2 \theta_p \cos^2 \theta_p \sin^2 \phi_p \right) \\ &= \beta'^4 \left(\sin^2 \theta_p \cos^2 \theta_p + \sin^2 \theta \cos^2 \theta + 2 \sin \theta_p \cos \theta_p \cos \phi_p \sin \theta \cos \theta \right). \end{aligned}$$

Using Eq. (15.4), we have

$$\frac{d^3 \sigma_1}{d(\hbar \omega) d\Omega d\Omega_p} = \frac{d^2 N_1}{d(\hbar \omega) d\Omega} \cdot \frac{d\sigma}{d\Omega_p} = \frac{d^2 N_1}{d(\hbar \omega) d\Omega} \cdot \frac{1}{4} R^2,$$

where

$$\frac{d\sigma}{d\Omega_p} = \frac{1}{4} R^2$$

is the classical hard sphere scattering cross section. Then,

$$\begin{aligned} \frac{d^2 \sigma_1}{d(\hbar \omega) d\Omega} &= \int \frac{d^2 N_1}{d(\hbar \omega) d\Omega} \cdot \frac{1}{4} R^2 d\Omega_p \\ &= \frac{q^2}{\hbar c} \frac{1}{\pi^2} \frac{1}{\hbar \omega} \int \beta'^4 \left(\sin^2 \theta_p \cos^2 \theta_p + \sin^2 \theta \cos^2 \theta + 2 \sin \theta_p \cos \theta_p \cos \phi_p \sin \theta \cos \theta \right) \\ &\quad \times \frac{1}{4} R^2 d\Omega_p \\ &= \frac{q^2}{\hbar c} \frac{\beta'^4 R^2}{4\pi^2} \frac{1}{\hbar \omega} \int d\phi_p \int d\theta_p \left(\sin^2 \theta_p \cos^2 \theta_p + \sin^2 \theta \cos^2 \theta \right) \sin \theta_p, \end{aligned}$$

where we have dropped the obviously 0 term. Perform the angular integration, we will finally get

$$\frac{d^2 \sigma_1}{d(\hbar \omega) d\Omega} = \frac{q^2}{\hbar c} \frac{\beta'^4 R^2}{4\pi^2} \frac{1}{\hbar \omega} \cdot 2\pi \cdot \left(\frac{4}{15} + 2 \sin^2 \theta \cos^2 \theta \right) = \frac{q^2}{\hbar c} \frac{\beta'^4 R^2}{3\pi} \frac{1}{\hbar \omega} \left(\frac{2}{5} + 3 \sin^2 \theta \cos^2 \theta \right).$$

(ii) For the second part, we can perform the calculation exactly as in Problem 15.2, where we have

$$\boldsymbol{\beta}'_i = \beta'(0, 0, 1), \quad \boldsymbol{\beta}'_f = \beta'(\sin \theta_p \cos \phi_p, \sin \theta_p \sin \phi_p, \cos \theta_p),$$

and

$$\Delta \boldsymbol{\beta}' = \beta'(\sin \theta_p \cos \phi_p, \sin \theta_p \sin \phi_p, \cos \theta_p - 1).$$

From this, we can also determine the point of impact as

$$\mathbf{r} = R \left(\cos \left(\frac{\theta_p}{2} \right) \cos \phi_p, \cos \left(\frac{\theta_p}{2} \right) \sin \phi_p, -\sin \left(\frac{\theta_p}{2} \right) \right),$$

Again,

$$\mathbf{n} = (\sin \theta, 0, \cos \theta), \quad \boldsymbol{\epsilon}_1 = (\cos \theta, 0, -\sin \theta), \quad \boldsymbol{\epsilon}_2 = (0, 1, 0),$$

then

$$\mathbf{n} \cdot \mathbf{r} = R \left(\sin \theta \cos \left(\frac{\theta_p}{2} \right) \cos \phi_p - \cos \theta \sin \left(\frac{\theta_p}{2} \right) \right).$$

To get the contribution from the second part, we need to calculate the following,

$$\sum_{i=1}^2 \left| \boldsymbol{\epsilon}_i \cdot \left(\mathbf{n} \times (\mathbf{n} \times \Delta \boldsymbol{\beta}') \right) (\mathbf{n} \cdot \mathbf{r}) \right|^2 = (\mathbf{n} \cdot \mathbf{r})^2 \cdot \sum_{i=1}^2 \left| \boldsymbol{\epsilon}_i \cdot \Delta \boldsymbol{\beta}' \right|^2.$$

Notice that

$$\sum_{i=1}^2 \left| \boldsymbol{\epsilon}_i \cdot \Delta \boldsymbol{\beta}' \right|^2 = \left| \Delta \boldsymbol{\beta}' \right|^2 - \left| \mathbf{n} \cdot \Delta \boldsymbol{\beta}' \right|^2 = \left| \mathbf{n} \times \Delta \boldsymbol{\beta}' \right|^2,$$

where

$$\mathbf{n} \times \Delta \boldsymbol{\beta}' = (-\cos \theta \sin \theta_p \sin \phi_p, \sin \theta(1 - \cos \theta_p) + \cos \theta \sin \theta_p \cos \phi_p, \sin \theta \sin \theta_p \sin \phi_p),$$

then we have

$$\begin{aligned} & \sum_{i=1}^2 \left| \boldsymbol{\epsilon}_i \cdot \left(\mathbf{n} \times (\mathbf{n} \times \Delta \boldsymbol{\beta}') \right) (\mathbf{n} \cdot \mathbf{r}) \right|^2 \\ &= \beta'^2 R^2 \left[\sin \theta \cos \left(\frac{\theta_p}{2} \right) \cos \phi_p - \cos \theta \sin \left(\frac{\theta_p}{2} \right) \right]^2 \\ & \quad \cdot \left[\cos^2 \theta \sin^2 \theta_p \sin^2 \phi_p + (\sin \theta(1 - \cos \theta_p) + \cos \theta \sin \theta_p \cos \phi_p)^2 + \sin^2 \theta \sin^2 \theta_p \sin^2 \phi_p \right]. \end{aligned}$$

Now,

$$\frac{d^3 \sigma_2}{d(\hbar \omega) d\Omega d\Omega_p} = \frac{d^2 N_2}{d(\hbar \omega) d\Omega} \cdot \frac{d\sigma}{d\Omega_p} = \frac{d^2 N_2}{d(\hbar \omega) d\Omega} \cdot \frac{1}{4} R^2,$$

where

$$\frac{d^2 N_2}{d(\hbar \omega) d\Omega} = \frac{q^2}{\hbar c} \frac{1}{\pi^2} \frac{1}{\hbar \omega} \left(\frac{\omega}{2c} \right)^2 \sum_{i=1}^2 \left| \boldsymbol{\epsilon}_i \cdot \left(\mathbf{n} \times (\mathbf{n} \times \Delta \boldsymbol{\beta}') \right) (\mathbf{n} \cdot \mathbf{r}) \right|^2.$$

Perform the integral (I used SymPy), we will get

$$\frac{d^2 \sigma_2}{d(\hbar \omega) d\Omega} = \frac{q^2}{\hbar c} \frac{1}{\pi^2} \frac{1}{\hbar \omega} \left(\frac{\omega}{2c} \right)^2 \frac{\beta'^2 R^4}{4} \frac{\pi}{6} (8 \cos^4 \theta + 16 \sin^2 \theta \cos^2 \theta + 5 \sin^4 \theta).$$

Rearrange terms, we will get

$$\frac{d^2\sigma_2}{d(\hbar\omega)d\Omega} = \frac{q^2}{\hbar c} \frac{\beta'^2 R^2}{12\pi} \frac{1}{\hbar\omega} \left(\frac{\omega R}{c}\right)^2 \left(1 - \frac{3}{8} \sin^4 \theta\right).$$

The total radiation cross section is then given by

$$\begin{aligned} \frac{d^2\sigma}{d(\hbar\omega)d\Omega} &= \frac{d^2\sigma_1}{d(\hbar\omega)d\Omega} + \frac{d^2\sigma_2}{d(\hbar\omega)d\Omega} \\ &= \frac{q^2}{\hbar c} \frac{\beta'^4 R^2}{3\pi} \frac{1}{\hbar\omega} \left(\frac{2}{5} + 3 \sin^2 \theta \cos^2 \theta\right) + \frac{q^2}{\hbar c} \frac{\beta'^2 R^2}{12\pi} \frac{1}{\hbar\omega} \left(\frac{\omega R}{c}\right)^2 \left(1 - \frac{3}{8} \sin^4 \theta\right). \end{aligned}$$

Replace β' with $\beta/2$, we finally get

$$\frac{d^2\sigma}{d(\hbar\omega)d\Omega} = \frac{q^2}{\hbar c} \frac{\beta^2 R^2}{48\pi} \frac{1}{\hbar\omega} \left[\beta^2 \left(\frac{2}{5} + 3 \sin^2 \theta \cos^2 \theta\right) + \left(\frac{\omega R}{c}\right)^2 \left(1 - \frac{3}{8} \sin^4 \theta\right) \right].$$

It is straightforward to verify that the total cross section is

$$\frac{d\sigma}{d(\hbar\omega)} = \frac{q^2}{\hbar c} \frac{\beta^2 R^2}{15} \frac{1}{\hbar\omega} \left[\beta^2 + \left(\frac{\omega R}{c}\right)^2 \right].$$